Codility_

CodeCheck Report: training6QCATV-VWG

Test Name:

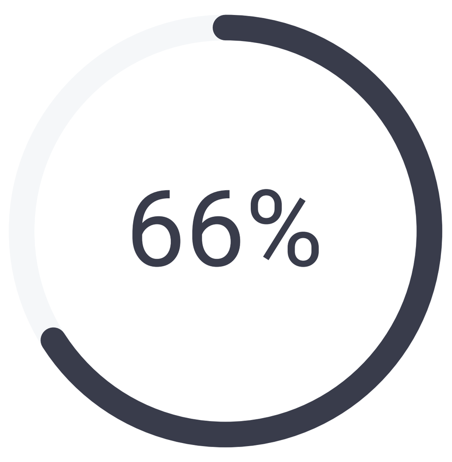
Check out Codility training tasks

Summary | Timeline

Tasks summary

Task		Time spent	Score
CountNonDivisible Python	⚠	2 min	66%

Total score



Tasks Details

Medium

1. CountNonDivisible

Calculate the number of elements of an array that are not divisors of each element.

Task Score	Correctness	Performance
66%	100%	25%

Task description

You are given an array A consisting of N integers.

For each number A[i] such that 0 ≤ i < N, we want to count the number of elements of the array that are not the divisors of A[i]. We say that these elements are non-divisors.

For example, consider integer N = 5 and array A such that:

```
A[0] = 3
A[1] = 1
A[2] = 2
A[3] = 3
A[4] = 6
```

For the following elements:

- A[0] = 3, the non-divisors are: 2, 6,
- A[1] = 1, the non-divisors are: 3, 2, 3, 6,
- A[2] = 2, the non-divisors are: 3, 3, 6,
- A[3] = 3, the non-divisors are: 2, 6,
- A[4] = 6, there aren't any non-divisors.

Write a function:

```
def solution(A)
```

that, given an array A consisting of N integers, returns a sequence of integers representing the amount of non-divisors.

Result array should be returned as an array of integers.

For example, given:

```
A[0] = 3
A[1] = 1
A[2] = 2
A[3] = 3
A[4] = 6
```

the function should return [2, 4, 3, 2, 0], as explained above.

Write an **efficient** algorithm for the following assumptions:

- N is an integer within the range [1..50,000];
- each element of array A is an integer within the range [1..2 ∗ N].

Solution

Programming language used:	Python
Total time used:	2 minutes
Effective time used:	2 minutes
Notes:	*not defined yet*

Task timeline

05:52:45 — 05:54:19

Code: 05:54:19 UTC, py, final, score: 66

show code in pop-up

```python
# you can write to stdout for debugging p
# print("this is a debug message")

def solution(A):
    N = len(A)

    # Count occurrences of each number
    count = {}
    for num in A:
        if num not in count:
            count[num] = 0
        count[num] += 1

    # Find divisors and count non-divisor
    result = []
    for num in A:
        divisors_count = 0
        # Check numbers from 1 to sqrt(nu
        j = 1
        while j * j <= num:
            if num % j == 0:
                # j is a divisor
                if j in count:
                    divisors_count += cou
                # num/j is also a divisor
                if j != num // j and num
                    divisors_count += cou
            j += 1
        result.append(N - divisors_count)

    return result
```

Analysis summary

The following issues have been detected: timeout errors.

Analysis

Detected time complexity: O(N ** 2)

expand all	Example tests
▶ example example test	✔ OK

expand all	Correctness tests
▶ extreme_simple extreme simple	✔ OK
▶ double two elements	✔ OK
▶ simple simple tests	✔ OK
▶ primes prime numbers	✔ OK
▶ small_random small, random numbers, length = 100	✔ OK

expand all	Performance tests
▶ medium_random medium, random numbers length = 5,000	✔ OK
▶ large_range 1, 2, ..., N, length = ~20,000	✘ TIMEOUT ERROR running time: 0.160 sec., time limit: 0.160 sec.
▶ large_random large, random numbers, length = ~30,000	✘ TIMEOUT ERROR running time: 0.596 sec., time limit: 0.320 sec.
▶ large_extreme large, all the same values, length = 50,000	✘ TIMEOUT ERROR running time: 1.836 sec., time limit: 0.272 sec.